University Institute of Engineering
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EXPERIMENT: 3

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SEMESTER: 6TH **SUBJECT CODE: 23CSH-314**
SUBJECT NAME: SYSTEM DESIGN

1. Aim:

To design a Social Media Platform similar to Facebook / Instagram
Description:
A social media platform is a platform which allows users to share photos, videos, and text with their friends and followers.
To design a scalable and highly available Social Media System where users can register, create posts, follow others, and interact with content through likes and comments

2. Objective:

- To understand the working of a large-scale social media system
- To design functional and non-functional requirements
- To identify core system entities
- To design API endpoints for communication
- To ensure scalability, availability, and low latency

3. Tools Used: Draw.io, Lucidchart, Excalidraw

4. Requirements Specification:

4.1. Functional Requirements:

- User should be able to register and login.
- User should be able to create posts (text / image / video).
- Users should be able to follow each other / send friend requests.
- Users should be able to like and comment on posts.
- Users should be able to view feed from people they follow

4.2. Non-Functional Requirements:

- Scalability:
  - System should support 500 million DAU.
- Consistency & Availability (CAP Theorem):
  - This system needs to be highly available first, then consistent.
  - Reason: If the system is not operational, it becomes useless.

    - Example:
    - If Instagram is down for 1 hour → huge issue.
    - But if a post takes 500ms to reach followers → acceptable.
    - Hence: Availability >>>Consistency
- Latency: Uploading / publishing post should take ≤ 500 ms.

4.3. Core entities of the system:

- User
- Post
- Comment
- Like
- Follow / Friend Request
- Feed
- Media (Image/Video)

5. API Design:

- User Registration API: POST /api/users/register
- User Login API: POST /api/users/login
- Create Post API: POST /api/posts
- Follow User API: POST /api/users/{user_id}/follow
- Like Post API: POST /api/posts/{post_id}/like
- Comment API: POST /api/posts/{post_id}/comment
- Get Feed API: GET /api/feed?limit=20

6. Database Schema:

The system is divided into domain-specific schemas to support microservices:

- Identity (MySQL): users table storing credentials and contact info.
- Catalog (MS SQL): products, categories, and sellers tables.
- Media (S3): product_images table linked to image URLs in S3.
- Cart & Inventory (PostgreSQL): cart, cart_items, and inventory (with reserved_quantity).
- Orders & Payment (MySQL): orders, order_items, and payments tables.
- Social: reviews table for user feedback and ratings.

7. High Level Design (HLD):

The system consists of Client, API Gateway, Backend Services (User, Content, Feed, Follower, Engagement), Message Queue (Kafka), Cache, and Databases. Media content is stored in object storage.

8. Learning Outcomes:

8.1. Designed a scalable Social Media system.
8.2. Identified functional and non-functional requirements.
8.3. Designed RESTful APIs for core operations.
8.4. Understood feed generation strategies.